

Server Architecture, Performance Audit & Capacity Report

Audit Date: August 28, 2026 | Instance: AWS EC2 t3.small (ap-south-2 Hyderabad) | Host: 40.192.34.67 | Database: PostgreSQL 16.15

Executive Summary: TechTrolley is architected as a lightweight **Single Page Application (SPA)** paired with a stateless Django 6.0 REST API and PostgreSQL 16. Client browsers execute searching, filtering, and printing locally, keeping server overhead minimal. Our live empirical audit confirms the current server is operating at **less than 20% of its real-world capacity**. The entire production database occupies just **16 MB**, and total media files occupy only **14 MB**. The system can easily handle **50 to 100 concurrent active users** for AM Audiovisuals, or **5 to 10 distinct companies** in a multi-tenant setup before needing any server upgrade.

1. Live EC2 Production Specifications & Empirical Metrics

Metric / Component	Specification	Current Usage	Available Headroom
Instance Type	AWS EC2 t3.small	Standard Baseline	Full CPU Burst Credits
vCPU / Processor	2 vCPUs (Intel Xeon Platinum 8259CL @ 2.50GHz)	Load Avg: 0.17 (~15% load)	~85% CPU Idle
Total System RAM	1.86 GiB (1,907 MB)	890 MB (46.6%)	1,017 MB (53.4%)
Swap Space	2,048 MB (Disk swap)	54 MB (2.6%)	1,994 MB Free
Disk Storage (EBS)	29.0 GB gp3 SSD	7.8 GB (28% used)	21.0 GB Free (72%)
Operating System	Ubuntu 24.04.4 LTS (Linux 6.17)	Uptime: 159 Days	Zero crash loops
PostgreSQL 16 DB	Local PostgreSQL 16.15 (port 5432)	16 MB total DB size	2,263 assets, 2,921 challan records
Media Storage (PDFs)	/media/conference_pdfs/	14 MB total media	> 20 GB free space

2. Memory (RAM) Breakdown: OS Overhead vs. Application

Layer / Service	Process / Daemons	RAM Used	% of Total RAM
OS & Kernel Services	systemd, journald, sshd, networkd, snapd, cron	~260 MB	13.6%
Gunicorn WSGI App	3 Django workers (~162 MB ea) + 1 master (26 MB)	~513 MB	26.9%
PostgreSQL 16 Engine	Shared memory buffers, WAL writer, background worker	~90 MB	4.7%
Nginx Web Proxy	SSL termination, reverse proxy to port 8000	~10 MB	0.5%
Free / Page Cache Buffer	Instantly reclaimable memory for query bursts	~1,017 MB	53.3%
Total System Capacity	AWS EC2 t3.small memory bank	1,907 MB	100.0%

Key Determination: The base Ubuntu 24 OS requires only ~260 MB. Gunicorn and PostgreSQL together consume ~600 MB. At 890 MB active memory, your system operates at 46.6% utilization with over 1 GB of buffer. A **2 GB RAM** host is the ideal baseline configuration.

3. User Capacity Estimation (Without Server Load)

- Because TechTrolley is a Single Page Application (SPA), searching equipment, filtering categories, calculating tax/rates, and assembling challans occur on the **user's device**. The server only delivers lightweight JSON data.
- **Concurrent Active Staff: 50 to 100 users** simultaneously scanning and updating challans.
 - **Registered Accounts: 1,000+ staff accounts** without performance impact.
 - **Current Workload vs. Ceiling:** AM Audiovisuals currently has 16 system users with 3 to 10 active at peak times. You are using **less than 10%** of the current server's capacity.

4. Multi-Company Expansion: Architectural Options & Resource Needs

If you want to onboard or sell this system to other AV rental companies:

Option A: Shared Multi-Tenant Database (Recommended for SaaS)

Each record is tagged with company_id. All companies share the single PostgreSQL instance and Django app. Resource cost is ~15–25 MB storage per company/year. Current t3.small can host **5 to 10 companies** (~150 active users) with **zero hardware upgrades**.

Option B: Isolated Database per Company (Best for Legal/Audit Separation)

Each company has its own PostgreSQL database (e.g. techtrolley_comp1) on the same host. Zero risk of cross-client data access. A 4 GB RAM host easily hosts **15 to 25 isolated company databases**.

Option C: Dedicated White-Label VPS per Company (For Enterprise Clients)

Deploy a dedicated lightweight \$5/month cloud VPS per client under their own custom domain and branding. Dedicated resources with complete physical isolation.

5. Sizing Recommendation Tiers & Cloud Cost Comparison

Scale Tier	Target Capacity	Hardware Recommendation	Est. Monthly Cost
Tier 1: Single Company (Current)	AM Audiovisuals (~10–30 users, ~5K assets)	1–2 vCPU, 2 GB RAM, 25 GB SSD	\$5 – \$15 / mo
Tier 2: Small Multi-Company	Up to 10 Companies (~150 users, ~30K assets)	2 vCPU, 4 GB RAM, 50 GB SSD	\$12 – \$30 / mo
Tier 3: Growing SaaS Platform	Up to 50 Companies (~500 users, ~150K assets)	4 vCPU, 8 GB RAM, 100 GB SSD	\$25 – \$60 / mo
Tier 4: Enterprise Scale	100+ Companies (> 1,500 active users)	4–8 vCPU, 16 GB RAM + RDS Postgres + S3	\$100 – \$200 / mo

Recommended Hosting Providers for TechTrolley

Provider	Plan	Specifications	Price	Key Advantage
Hetzner Cloud	CX22 / CPX21	2–3 vCPU, 4 GB RAM, 40–80 GB NVMe	€4.50–€7.50 (~\$5–\$8)	Highest performance per dollar globally
AWS EC2 (Current)	t3.small	2 vCPU, 2 GB RAM, 30 GB EBS	~\$15 – \$20	AWS ecosystem, already configured
DigitalOcean	Basic Droplet	1–2 vCPU, 2 GB RAM, 50 GB SSD	~\$12 – \$18	1-click automated backups & easiest UI
Linode (Akamai)	Shared 2GB/4GB	1–2 vCPU, 2–4 GB RAM, 50–80 GB SSD	~\$12 – \$24	Excellent Mumbai/India datacenter latency